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Compiler Construction

Assignment 1

1- Language Overview

Language Name: C++ : Bankai Mode

Theme: Bankai is a famous word from Japanese anime Bleach and its meaning is "unleash it to its ultimate form." So by C++ : Bankai Mode it means "where every line of code unleashes your ultimate ability."

In C++ Bankai Mode, every keyword channels its final form by adding `_bankai`.

The language blends traditional C++ logic with the epic intensity of anime and each statement feels like shouting a battle cry!

Purpose:

To make programming fun and expressive, letting developers feel like anime warriors commanding digital powers instead of just typing syntax.

Example Program

```
void_bankai main_bankai() {  
    int_bankai power = 9000;  
    if_bankai (power > 8000)  
    {  
        display_bankai << "It's over 9000!!!";  
    }  
    return_bankai 0;  
}
```

2- Keywords

Key word	Meaning / Equivalent in C++	Power / Reference	Example usage
1- int_bankai	int	Power level variable	int_bankai aura = 100;
2- float_bankai	float	Fluid energy variable	float_bankai speed = 7.9;
3- char_bankai	char	Single rune of power	char_bankai rank = 'S';
4- String_bankai	string	phrase spell	string_bankai move = "Cut";
5- void_bankai	void	Energy void function	void_bankai summon(1) {};
6- if_bankai	if	Battle condition	if_bankai (health < 50)
7- else_bankai	else	Alternate timeline	else_bankai { heal(1); }
8- display_bankai	cout	Show power	display_bankai << fire;
9- cin_bankai	cin	take power	cin_bankai >> x;
10- for_bankai	for	training arc loop	for_bankai (i=0; i<10; i++)
11- while_bankai	while	infinite battle mode	while_bankai (true)
12- return_bankai	return	Power discharge	return_bankai 0;
13- continue _bankai	continue	retry mode	continue_bankai;
14- break_bankai	break	Escape technique	break_bankai;
15- main_bankai	main()	Main technique	main_bankai ();

Operators and Punctuations

Category	Symbol	Description
Operator	add +	performs addition
Operator	sub -	performs subtraction
Operator	mul *	performs multiplication
Operator	div /	performs division
Operator	rem %	finds remainder of div
Operator	==	Compare or checks equality
Punctuation	;	Statement terminator
Punctuation	,	Separator for list items or func parameters
Punctuation	{	opening of code block
Punctuation	}	closing of code block

4- Regular Expression (RE)

Token	RE	Example
integers	$[+ -]? [0-9]^+$	+11 -1 2
float	$[+ -]? [0-9]^+ \backslash . [0-9]^+$	3.14 +1.6 -0.99
identifiers	$[a-zA-Z]^+ ([a-zA-Z0-9_])^* [a-zA-Z0-9_]?$	a a1 a-1 A9
Keywords	$[int_bunkai \mid float_bunkai \mid char_bunkai \mid$ $String_bunkai \mid void_bunkai \mid if_bunkai \mid$ $else_bunkai \mid display_bunkai \mid cin_bunkai \mid$ $for_bunkai \mid while_bunkai \mid return_bunkai \mid$ $continue_bunkai \mid break_bunkai \mid main_bunkai]$	
	all keywords eg int_bunkai or return_bunkai etc	

Token RE

Operators $[add+ | sub- | mul* | div/ | rem\% | ==]$

Example

$a + b$
 $a == b$

(a and b are just for example)

Punctuation $[; | , | \{ | \} |]$

$a; \{b \dots\}$

Comments $\# [^ \# \backslash n]^* \#$

✓ $\#$ This is a car $\#$

X $\#$ This

String literal $" ([^ " \backslash \backslash] | \backslash \backslash .)^* "$

✓ "Hello Bahkai"

✓ "She said 'hi' loudly"

X "Ali"

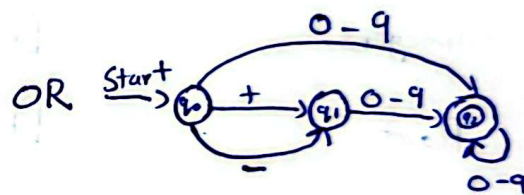
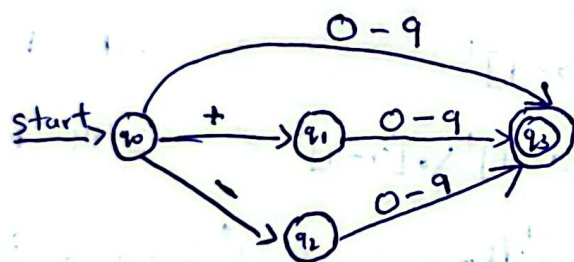
Character literal $'[^']'$

'A' 'a' '1'

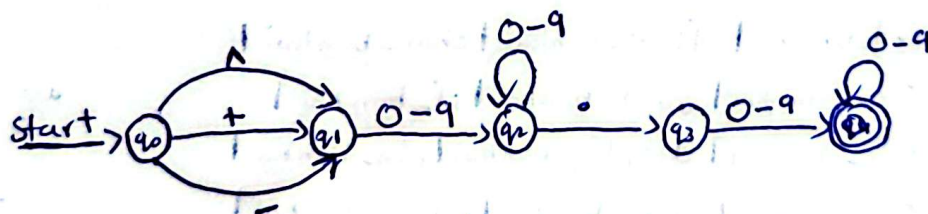
White space

5. Finite Automata Diagrams

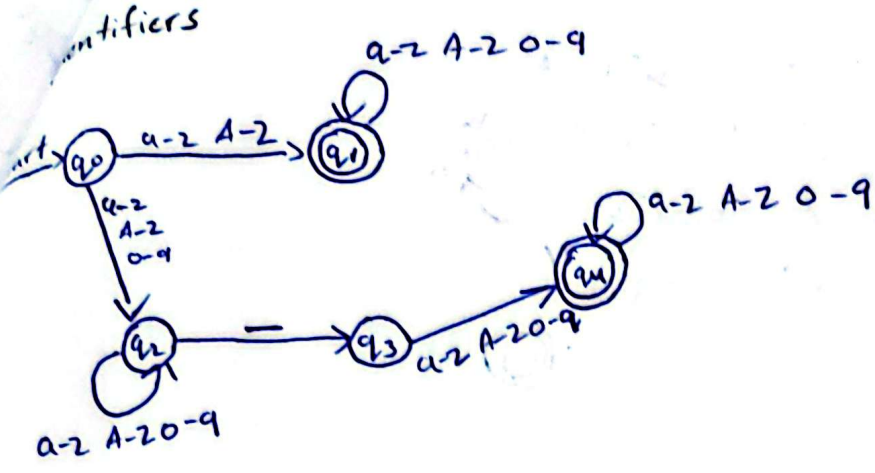
1- Integers



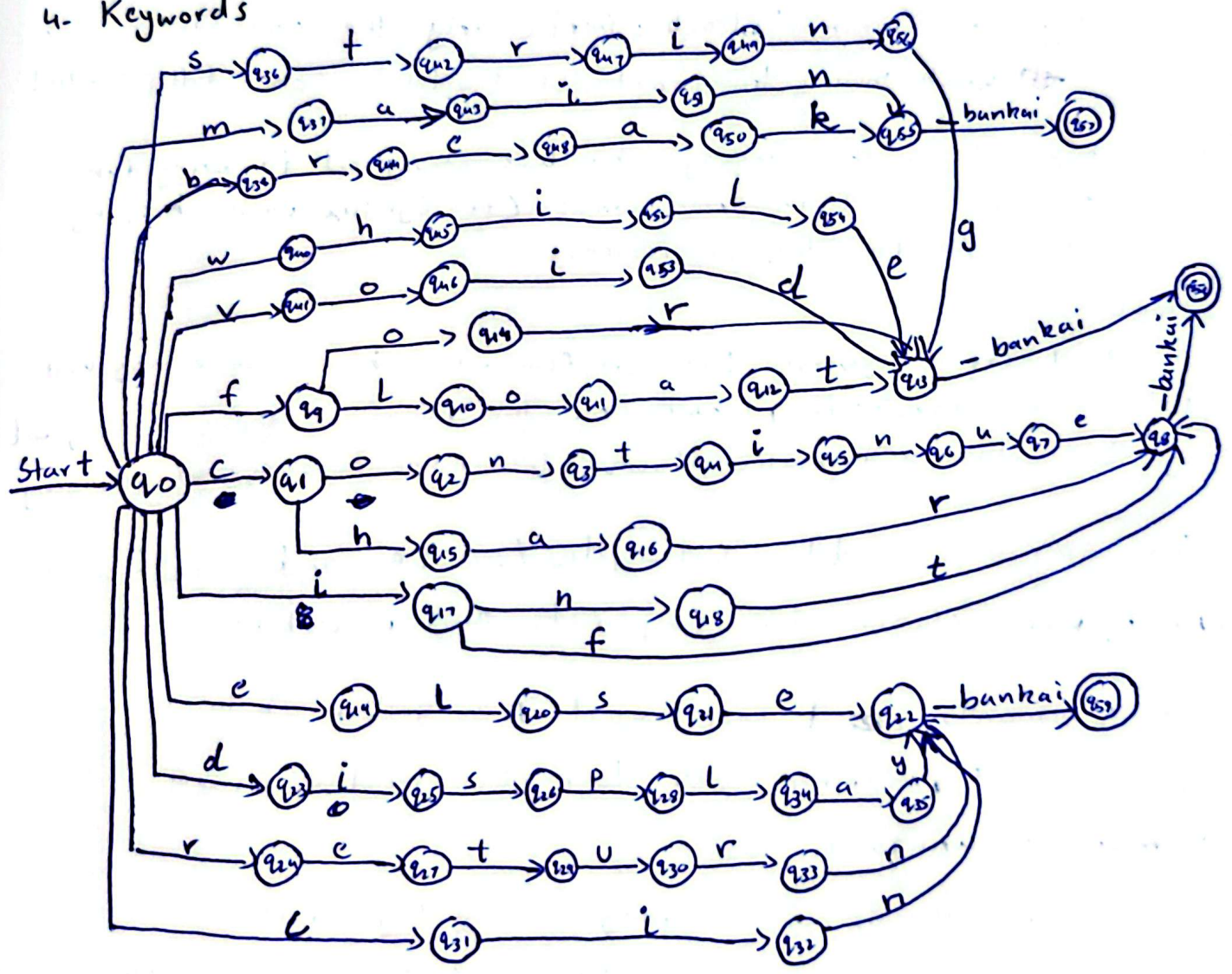
2- Float



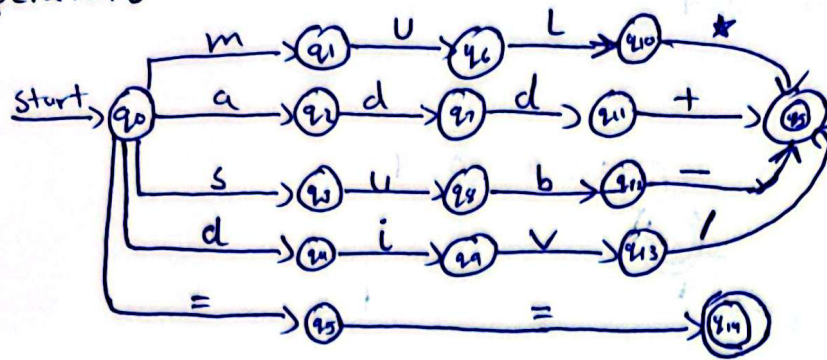
id + b
6
st for example
word



4. Keywords



5- Operators



6- Explanation

The C++ Bankai Mode is inspired by the powerful and dramatic style of anime "Bleach", particularly the concept of "Bankai" which means transformation that unleashes one's full potential.

Each keyword in this language has been designed to bring fun and expressive twist in traditional C++ syntax while keeping the structure similar for programmers.

For example

Standard C++ keywords like `int`, `float`, `char` have been reimagined as `int_bankai`, `char_bankai`, `if_bankai` etc. This maintains the same logical meaning but adds a thematic flavour that makes coding more engaging.

The operators `add+`, `sub-`, `div/`, `mul*` and punctuation `{`, `}` were chosen for simplicity and readability.

For comments, I have used `#` symbol both at start and end, making code look clean and distinct. What makes it unique?

It's the fusion of programming logic with creativity and personality. It embraces the energy and symbolism of anime, while also carrying simplified version of C++